A2 Notes FCP, Permutations and Combinations Day 2

Warm Up

Combination - select a group, treat the same   Permutation - select and treat differently

1: A club has 9 members. In how many ways can a president, vice president and secretary be elected from the members of the club?

treated differently ⇒ Permutation

$_9P_3 = 504$

2: A group of seven students working on a project needs to choose two from their group to present the group's report to the class. How many ways can they choose the two students?

→ combinations

$_7C_2 = 21$

3: ~~A softball team has 10 players. How many batting orders are possible if everyone gets to bat?~~

7 factorial = $7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$

4. Kaylee is trying to pick out an outfit for the first day of school. She can choose from 7 pairs of pants, 6 t-shirts, 7 sweaters and 2 pairs of shoes. How many different outfits does Kaylee have to choose from?

FCP: $\underline{7} \cdot \underline{6} \cdot \underline{7} \cdot \underline{2} = 588$

Word Arrangements: a representation of permutation with repeated elements

Ex How many ways can the 10 letter word **repetition** be arranged?

↑↑ look identical

Take the permutation of the number of letters and divide by the permutation of any letters that repeat.

$= \dfrac{\#\text{ letters}!}{\text{repeat}! \cdot \text{repeat}!}$ $\dfrac{10!}{2! \cdot 2! \cdot 2!}$ 2 e's, 2 t's, 2 i's

→ if easy simplify first $\dfrac{10!}{8} = 453,600$

You! How many ways can the letters in the word **Mississippi** be arranged?

1) find total # letters → 11 letters
2) find repetitions
3) set up fraction

4 i's, 4 s's, 2 p's

$\dfrac{11!}{4! \cdot 4! \cdot 2!}$

calculator

$11! \div (4! \cdot 4! \cdot 4!) = 34,650$

All kinds of problems! No two problems are exactly alike - you have to think about each situation and then see it up.

Imagine yourself in the situation

1. A multiple choice test contains 10 questions. There are four possible answers for each question. In how many ways can a student answer the questions on the test if the student answers every question?

$\underset{\#1}{\underline{4}} \cdot \underset{\#2}{\underline{4}} \cdot \underline{4} \cdot \underline{4} \cdot \underline{4} \cdot \underline{4} \cdot \underline{4} \cdot \underline{4} \cdot \underline{4} \cdot \underline{4}$

repeated multiplication ⇒ exponents

$= (4^{10}) = \boxed{1{,}048{,}576}$

2. A Club has 2 sophomores and 5 juniors. They want to take a yearbook photo with a sophomore on the far left and a junior on the far right. How many ways can the line-up for the photo?

TOTAL of 7 students

Deal with restrictions first

2 soph
5 juniors

choices: $\underline{2} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} \cdot \underline{5} =$

1 soph
4 juniors } 5 students

3. A committee must be formed with 4 teachers and 5 students. If there are 7 teachers to choose from, and 15 students, how many different ways could the committee be made?

Big Picture

$\underset{\text{Choose teachers}}{\underline{{}_7C_4}} \cdot \underset{\text{choose students}}{\underline{{}_{15}C_5}} \boxed{=}$ 105,105 ways

4. In a running competition, a bronze, silver and gold medal must be given to the top three girls and top three boys. If 11 boys and 14 girls are competing, how many different ways could the six medals possibly be given out?

YOU

$\underset{\text{girls}}{\underline{{}_{14}P_3}} \cdot \underset{\text{boys}}{\underline{{}_{11}P_3}} = 2{,}162{,}160$

5. How many four digit lock codes can be formed for a lock if the first digit must be a $\boxed{5 \text{ or } 7}$ and the last digit must be 9? All other digits can be any number and repeated.

Repeating OK

$\underline{2} \cdot \underline{10} \cdot \underline{10} \cdot \underline{1}$

10 digits total
0 thru 9

No Repeating

$\underline{2} \cdot \underline{8} \cdot \underline{7} \cdot \underline{1}$

a digit ... a digit

$10 - 2 = 8$ digits left